

OPTIMIZING LOGISTICS THROUGH DATA ANALYSTICS: A CASE STUDY OF E.T. TRANSPORT

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Introduction

This report presents an in-depth analysis of E.T. Transportation's logistics operations with the goal of improving operational efficiency through data analytics. By utilizing Power BI for data visualization, this project aims to identify key performance indicators (KPIs) that will drive improvements in delivery times, fuel efficiency, vehicle utilization, and overall customer satisfaction. The report covers the step-by-step process of developing a data dashboard and the key insights derived from analyzing the logistics data of E.T. Transportation.

Company Overview

E.T. Transportation is a leading logistics provider in Canada, founded in 2005. The company specializes in Full Truck Load (FTL) and Less Than Truck Load (LTL) services, operating across Canada and the U.S. It manages a fleet of over 100 trucks and 200 trailers, including specialized equipment like Dry Vans, Flatbeds, and Roll Tite trailers to handle diverse cargo types, including hazardous materials. E.T. Transportation is dedicated to delivering efficient, data-driven logistics solutions that prioritize safety, efficiency, and customer satisfaction.

Objective

The main goal of this project is to use data analytics to enhance E.T. Transportation's operational efficiency. This involves reducing fleet downtime, improving delivery times, optimizing fuel consumption, and ultimately increasing customer satisfaction through data-driven strategies. By analyzing detailed logistics data, the project seeks to identify actionable insights that will lead to operational improvements and cost savings.

Dataset Overview

The data used for this project includes various aspects of E.T. Transportation's operations:

- **Shipments & Routes:** Detailed information on shipments and their respective routes.
- **Fuel Costs & Vehicle Utilization:** Data on fuel consumption per mile and vehicle usage.
- **Delivery Performance:** On-time delivery rates, traffic conditions, and causes of delays.

Dashboard Creation

5.1 Data Import and Cleaning

The first step was importing the logistics dataset into Power BI. The raw data was then cleaned to ensure accuracy and standardization, which involved correcting any errors, removing duplicates, and filling missing values to create a reliable foundation for analysis.

5.2 Data Modeling and Relationships

Once the data was clean, relationships between different tables (shipments, vehicles, costs, and routes) were modeled. This allowed for a relational analysis, enabling the discovery of meaningful insights across multiple variables.

5.3 Identifying Key Performance Indicators (KPIs)

The following KPIs were identified as crucial for measuring E.T. Transportation's performance:

1. **Fuel Efficiency:** Actual vs. estimated fuel consumption per mile.
2. **On-Time Delivery Rate:** Percentage of deliveries made on time.
3. **Vehicle Utilization:** Utilization rates of trucks, vans, and other vehicle types.

5.4 Visualization and Dashboards

Using Power BI, the KPIs were visualized through various charts and summary cards:

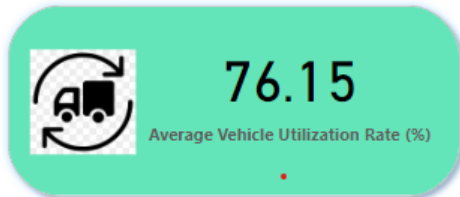
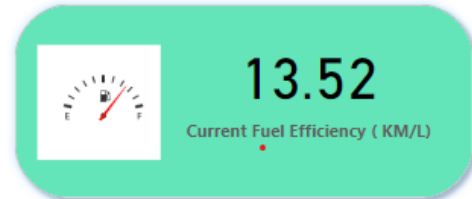
1. Bar Chart: Idle Hours for each Vehicle & Average and On-Time Delivery Time by Routes
2. Pie charts showed the proportion of on-time deliveries versus delays. These visualizations formed the basis of an interactive dashboard that provided a real-time snapshot of the company's logistics performance.

Key performance Index and Analysis

1. Current Fuel Efficiency (KM/L)

Value: 13.52 KM/L

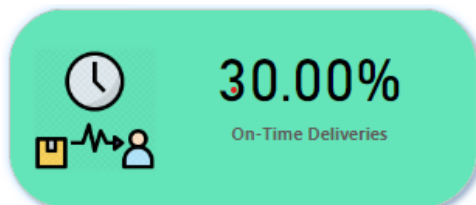
Analysis: This metric reflects the overall fuel efficiency across the fleet, indicating how many kilometers each vehicle can travel per liter of fuel. A value of 13.52 is decent for logistics operations but may require improvements in high-traffic routes or poorly performing vehicles.



2. Top Middle: Average Vehicle Utilization Rate (%)

Value: 76.15%

Analysis: This is the percentage of time that the fleet's vehicles are actively being used. A utilization rate of 76.15% suggests that there is some idle time, meaning about 24% of the fleet may be underutilized. Optimizing vehicle schedules could help boost this percentage, leading to better resource use.



3. On-Time Deliveries (%)

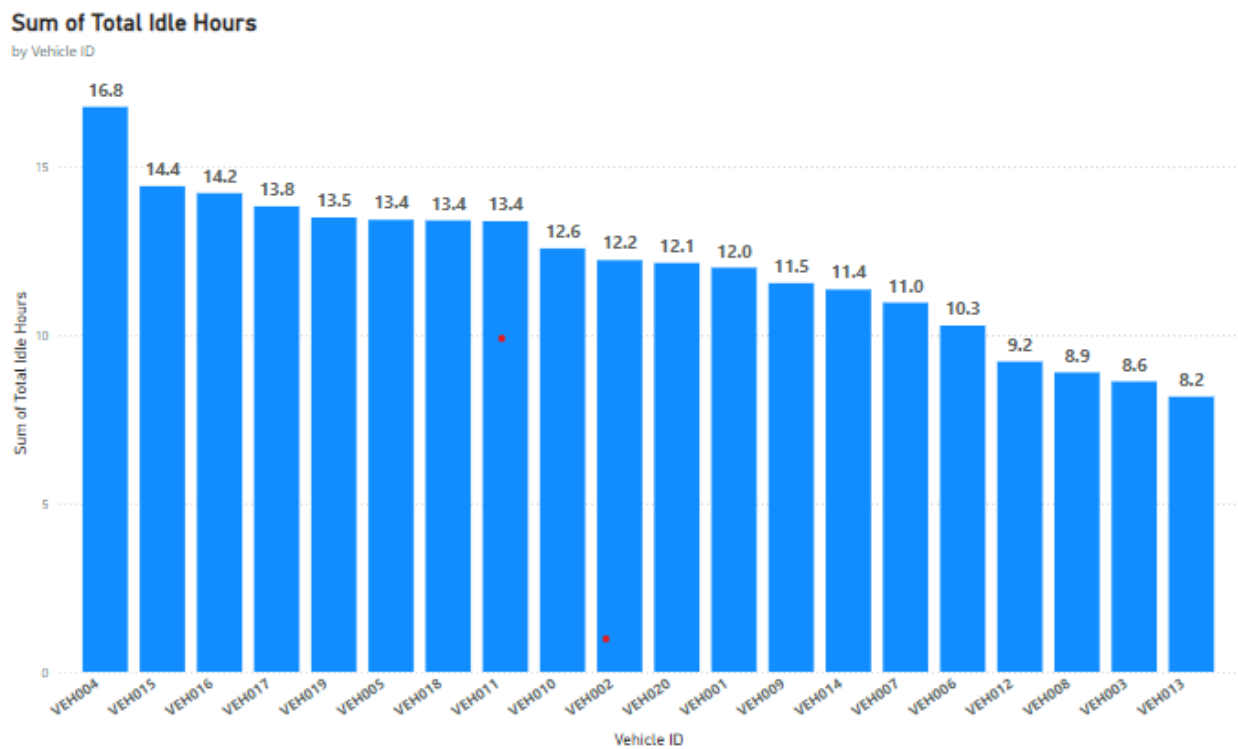
Value: 30.00%

Analysis: This is a concerning metric, with only 30% of deliveries arriving on time. This could indicate major inefficiencies in scheduling, route planning, or vehicle reliability. Improving this rate is crucial for customer satisfaction and operational efficiency.

4.Sum of Total Idle Hours by Vehicle ID

Graph Type: Bar Chart

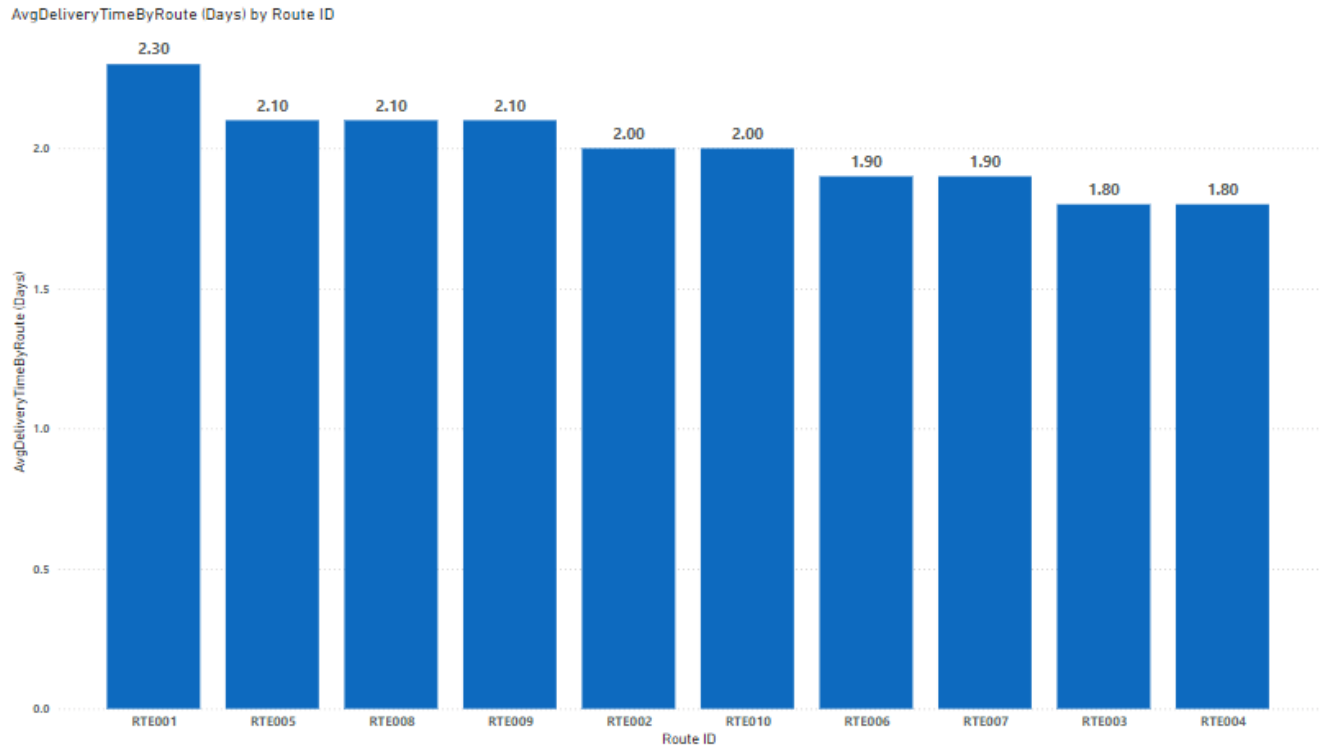
Analysis: The bar chart shows the idle hours per vehicle. Vehicle ID 98 has the highest idle time (16.8 hours), followed by several others with significant idle hours. The lowest idle time is Vehicle ID 73 (8.2 hours). Reducing idle time by reassigning underutilized vehicles or improving scheduling could enhance overall fleet



utilization.

5. Average Delivery Time in Days by Route ID

Graph Type: Bar Chart

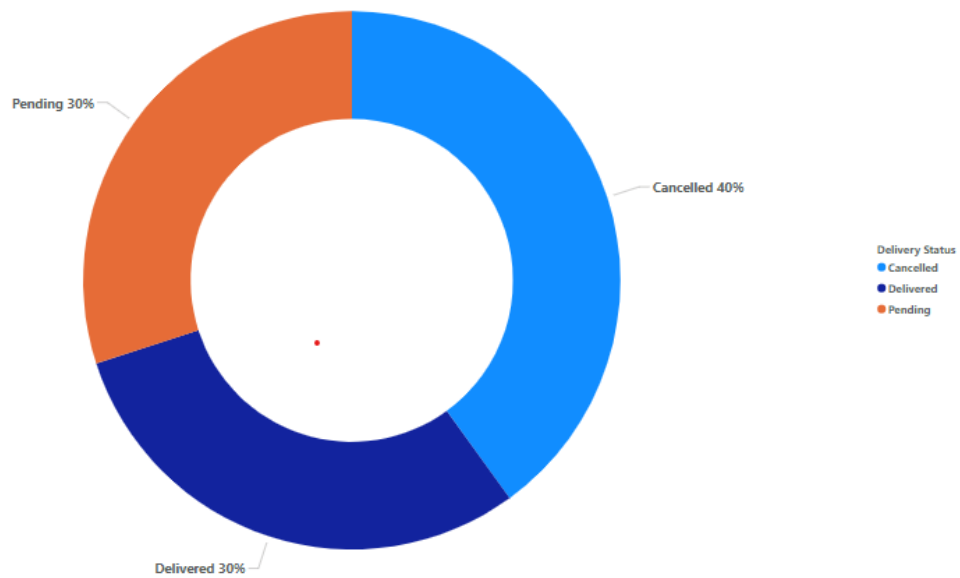


Analysis: This chart compares the average delivery time per route. Route RTE001 has the highest average delivery time (just over 2 hours), while RTE004 and RTE003 have shorter times. Identifying factors contributing to longer delivery times on certain routes, such as traffic or route length, can help optimize route planning and improve efficiency.

6. Delivery Percentage by Delivery Status

Graph Type: Donut Chart

Count of Delivery ID by Delivery Status



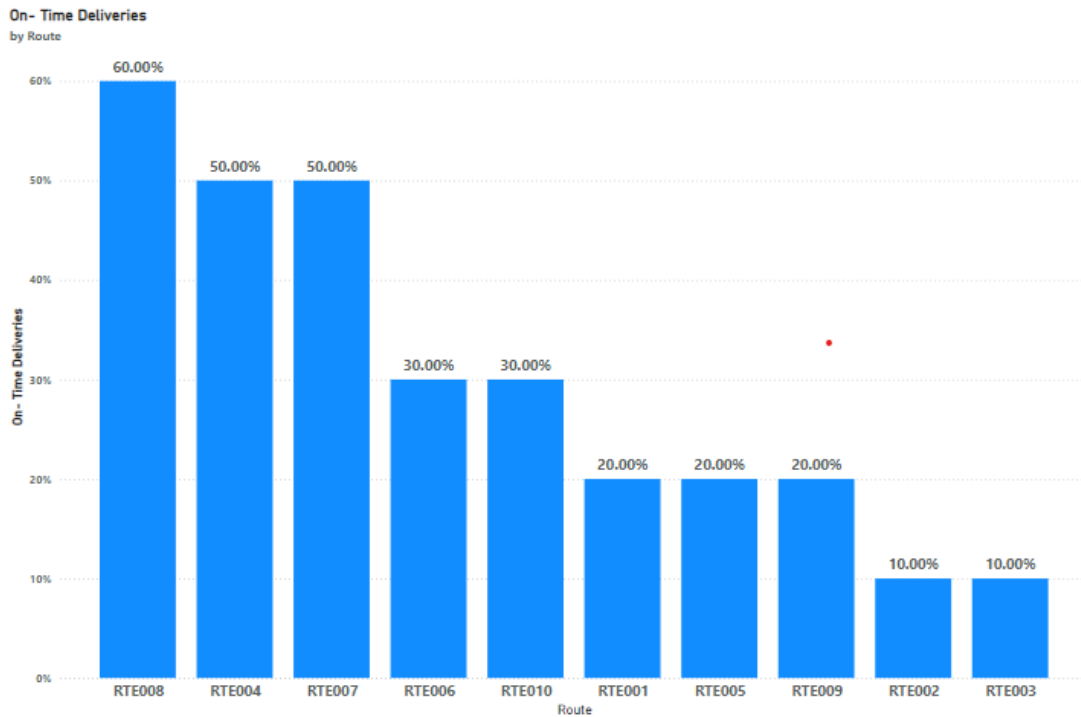
Analysis: The delivery statuses are broken down as:

- **Cancelled Deliveries:** 40%
- **Pending Deliveries:** 30%
- **Delivered Deliveries:** 30%

This shows that only 30% of deliveries are successfully completed, while the majority are either pending or cancelled. This points to significant operational issues, such as cancellations or delays, that need immediate attention to improve service reliability.

7. On-Time Deliveries by Routes

Graph Type: Bar Chart



Analysis: The percentage of on-time deliveries is broken down by route. Route RTE008 shows the highest percentage of on-time deliveries (around 55%), while routes such as RTE002, RTE003, and RTE001 have very low on-time delivery rates. This highlights the need to improve route planning or vehicle allocation on the underperforming routes to boost overall on-time performance.

Key Insights & Recommendations

- **Fuel Efficiency:** It's relatively good but could be optimized further.
- **Vehicle Utilization:** Slightly underutilized at 76%, and reducing idle time could help increase it.
- **On-Time Deliveries:** Significantly low at 30%, which is a major issue.
- **High Idle Hours:** Certain vehicles are showing high idle time, indicating possible inefficiencies in scheduling.
- **Delivery Status:** A large portion of deliveries are either cancelled or pending, which requires investigation into underlying causes.

- **Route-Specific Performance:** Some routes are significantly underperforming in terms of on-time deliveries and delivery time, calling for route optimization.

These insights can be used to create targeted strategies for improving efficiency, vehicle utilization, and on-time delivery rates across the logistics operations.

Conclusion

The findings in this report demonstrate that E.T. Transportation has significant potential to improve its operational efficiency through data-driven strategies. By optimizing fuel consumption, improving delivery times, and addressing vehicle utilization issues, the company can achieve considerable cost savings while enhancing customer satisfaction. The use of data analytics has provided clear, actionable insights for making informed business decisions. Going forward, it is recommended that E.T. Transportation continue refining its data analytics processes to stay competitive in the logistics industry.

Appendix

Dashboard

